BAIO LEARNING SYSTEM

Narendra Sir (Biology Lecturer, Founder & CEO)

HIGH-YIELD DIAGRAMS: PRIORITY & MARKS BLUEPRINT

Chapter: Molecular Basis of Inheritance • Ranked by Recurring Weightage, Crucial Labels & Marking Schemes

PRIORITY 1: ESSENTIAL 5-MARK SCHEMATIC DIAGRAMS

Frequently Tested • High Marking Weightage

1. Watson & Crick DNA Double Helix Structure

TOP PRIORITY | 5 MARKS

- **Diagram Concept:** Double-stranded right-handed helical chain with antiparallel polarity and base pairing.
- **Key Features to Render:** Major groove (22 Å), Minor groove (12 Å), Helical diameter (20 Å / 2.0 nm), One full turn pitch (34 Å / 3.4 nm), Axial distance between base pairs (3.4 Å / 0.34 nm).

Mandatory Labels: 5' and 3' ends (antiparallel), Sugar-Phosphate backbone, Adenine=Thymine (2 H-bonds), Guanine≡Cytosine (3 H-bonds), Major Groove, Minor Groove, Pitch (3.4 nm), Diameter (2.0 nm).

Marking Distribution: Neat Double Helix Outline: 2M | Dimensional Metrics (Pitch, Diameter, Rise): 1.5M | Correct Base-Pair Hydrogen Bonding & Polarity Labels: 1.5M.

2. Lac Operon (In Absence & Presence of Inducer)

TOP PRIORITY | 5 MARKS

- **Diagram Concept:** Two-state operon cassette diagram showing negative regulation vs. inducer-driven transcription.
- **State 1 (Off State - Absence of Lactose):** *i* gene → Repressor mRNA → Active Repressor binds Operator (o) → Blocks RNA Polymerase.
- **State 2 (On State - Presence of Lactose):** Inducer (Lactose/Allolactose) + Repressor → Inactive Repressor → Operator free → Polycistronic mRNA transcribed → Translation of β-galactosidase (z), Permease (y), Transacetylase (a).

Mandatory Labels: Promoter (p), Regulator (i), Operator (o), Structural genes (z, y, a), Repressor protein, Inducer (lactose), Inactive repressor complex, β-galactosidase, Permease, Transacetylase.

Marking Distribution: Switched OFF Diagram & Labels: 2.5M | Switched ON Diagram & Enzyme Translation Labels: 2.5M.

3. Hershey-Chase Experiment (Blender Experiment Flowchart)

TOP PRIORITY | 5 MARKS

- **Diagram Concept:** Two parallel infection pathways depicting bacteriophages labeled with ^{35}S (protein coat) vs ^{32}P (DNA core).
- **3-Step Sequential Flow:** 1. Infection → 2. Blending → 3. Centrifugation.

Mandatory Labels: Radioactive ^{35}S labeled protein capsule, Radioactive ^{32}P labeled DNA, *E. coli* cell, Infection, Blending, Centrifugation, Radioactive supernatant (^{35}S), Radioactive bacterial pellet (^{32}P).

Marking Distribution: ^{35}S Experimental Track & Result: 2M | ^{32}P Experimental Track & Result: 2M | 3-Step Labels (Infection, Blending, Centrifugation): 1M.

PRIORITY 2: CORE 3-MARK STRUCTURAL & PROCESS DIAGRAMS

Frequent 3-Mark Value Questions

4. Nucleosome Core Particle & 'Beads-on-a-String'

HIGH PRIORITY | 3 MARKS

- **Diagram Concept:** Negatively charged DNA double helix wound ~1.65 turns around a positively charged basic histone octamer.

Mandatory Labels: Histone Octamer Core (H2A, H2B, H3, H4), Core DNA (~200 bp), H1 Linker Histone.

Marking Distribution: Histone Octamer & Wrapped DNA Outline: 1.5M | H1 Histone & Core DNA Labels: 1.5M.

5. DNA Replication Fork (Continuous & Discontinuous Synthesis)

HIGH PRIORITY | 3 MARKS

- **Diagram Concept:** Y-shaped replication fork showing continuous leading strand and discontinuous Okazaki lagging fragments.

Mandatory Labels: Template parental strands (3'→5' and 5'→3'), Leading / Continuous strand (5'→3'), Lagging / Discontinuous strand, Okazaki fragments, Replication fork direction.

Marking Distribution: Y-fork Architecture & Strand Polarities: 1.5M | Continuous vs Okazaki Fragment Identification: 1.5M.

6. Schematic Structure of a Transcription Unit

HIGH PRIORITY | 3 MARKS

- **Diagram Concept:** Linear double-stranded DNA segment demarcating the structural gene flanked by regulatory boundaries.

Mandatory Labels: Promoter (5' upstream of coding strand), Structural Gene, Terminator (3' downstream of coding strand), Template Strand (3'→5'), Coding Strand (5'→3'), Transcription start site.

Marking Distribution: Three Functional Regions (Promoter, Structural Gene, Terminator): 1.5M | Template vs Coding Polarities: 1.5M.

7. Cloverleaf Structure of tRNA (The Adapter Molecule)

HIGH PRIORITY | 3 MARKS

- **Diagram Concept:** 2D cloverleaf showing base-paired stems and looped regions with codon-anticodon interaction.

Mandatory Labels: Amino acid acceptor end (3'-OH with CCA sequence), Anticodon loop (with complementary bases to mRNA codon), DHU loop, TψC loop, Variable loop, mRNA codon strand (5'→3').

Marking Distribution: Cloverleaf 4-Arm Structure: 1.5M | 3'-Acceptor & Anticodon Loop Labels: 1.5M.

PRIORITY 3: SCHEMATIC FLOWCHARTS & PROCESS PROOFS (2M / 3M)

Mechanism & Separation Profiles

8. Meselson & Stahl Experiment (CsCl Centrifugation Density Bands)

MEDIUM PRIORITY | 2M / 3M

Mandatory Labels: Heavy DNA band (^{15}N-^{15}N, bottom), Generation I Hybrid band (^{15}N-^{14}N, intermediate, 20 min), Generation II Light (^{14}N-^{14}N) and Hybrid bands (40 min), Centrifugation direction.

9. Eukaryotic Transcription Processing (hnRNA to mRNA)

MEDIUM PRIORITY | 2M / 3M

Mandatory Labels: hnRNA primary transcript, Exons (coding) & Introns (non-coding), Splicing (spliceosome loop excision), 5' Capping (m^7Gppp), 3' Poly-A tail (polyadenylation).

10. DNA Fingerprinting Banding Comparison (Gel Autoradiogram)

MEDIUM PRIORITY | 2M / 3M

Mandatory Labels: Paternal & Maternal Chromosomes (e.g., Chr 7, 2, 16), Variable Number of Tandem Repeats (VNTR copy numbers), Size-separated Agarose Gel lanes, Crime Scene Sample (C) matched with Suspect (B).